UAV Design & Simulation Internship - Task 4

Part 1: Understanding GPS-based Navigation

- Introduction to GPS-

→ GPS (Global Positioning System) is a satellite-based navigation system that allows a drone/object to determine its exact position anywhere on Earth.

→ It works using signals from multiple satellites orbiting the Earth.

- Components of GPS- Space Segment, Control Segment, User Segment
- GPS Coordinates System-

→ A location is defined using 3 values:

a) Latitude- Measures North-South position
-Range: -90° to +90°

b) Longitude - Measures East-West position
- Range: -180° to +180°

c) Altitude- Height from sea level (in meters)

- How GPS Works-

1. Satellites send signals with: Time, Position
2. Drone receives signals from at least 4 satellites, Using a method called ***Trilateration***
3. Drone calculates distance from satellites
4. Determines its exact 3D position

- UAV Navigation Concepts-

→ <u>Waypoint Navigation</u>
Drone moves from one coordinate to another
Used in missions like mapping, delivery

→ Path Planning
Deciding the best route between points

→ Position Hold
Drone stays at one fixed location

→ Return-to-Launch (RTL)
Drone automatically returns to starting point

- Importance of GPS in UAVs - Stable flight
  - Autonomous missions
  - Accurate positioning
  - Navigation in large areas
- Real-Life Applications - Agriculture (crop monitoring), Surveillance, Parcel delivery, Disaster management
- Limitations of GPS - Weak signal indoors
  - Errors due to weather or obstacles
  - Accuracy depends on satellite visibility
- Conclusion - GPS is the backbone of modern navigation. By using satellite signals and trilateration, drones/object can determine their position and execute given missions efficiently.

Part 2: Waypoint Mission Planning(QGroundControl)

- What is a Waypoint Mission?

→A waypoint mission is a set of GPS coordinates that the drone follows automatically in sequence.

- Steps to Create a Mission -

Step 1: Open Plan Mode
Launch QGroundControl
Click on "Plan"
Step 2: Select Location
Choose area on map
Step 3: Add Waypoints
Click on map to add points
Each click creates a waypoint
Step 4: Set Parameters - For each waypoint: Altitude, Speed (optional), Actions
Step 5: Add Special Commands- Takeoff, Land, Return-to-Launch (RTL)
Step 6: Save & Upload- Save mission & Upload to drone

- List of Coordinates used (Square Path) -

WP1 → 25°21'07.2"N 85°26'32.3"E → 10m
WP2 → 25°21'06.2"N 85°26'34.2"E → 15m
WP3 → 25°21'07.3"N 85°26'34.6"E → 20m
WP4 → 25°21'07.8"N 85°26'33.7"E → 12m

- Steps to run mission:

1. Connect drone (PX4 SITL or real drone)
2. Wait for GPS lock
3. Upload mission
4. Arm the drone
5. Switch to Auto Mode
6. Start mission

- Observations -

→Drone follows all waypoints sequentially
→Smooth transitions between point
→Altitude changes correctly

→RTL executed successfully

- Conclusion - QGroundControl simplifies UAV mission planning by providing a graphical interface to create and execute waypoint-based autonomous missions. It is widely used in both simulation and real-world drone operations.

Part3 : Autonoumous Flight Execution

- Introduction- Autonomous flight execution refers to the process where a UAV follows a pre-defined mission without manual control. This is achieved using waypoint navigation through an autopilot system like PX4, controlled via QGroundControl.
- Objective- To execute an autonomous waypoint mission using PX4 simulation and observe drone behavior such as navigation accuracy, altitude changes, and return-to-launch.
- Tools Used- PX4 SITL (Software-In-The-Loop), JMAVSim / Gazebo Simulator, QGroundControl
- Simulation Setup -
  Step 1: Launch PX4
  Step 2: Connect Ground Station
- Mission Execution Steps -
  Step 1: Upload Mission
  Step 2: Pre-Flight Checks
  Step 3: Arm the Drone
  Step 4: Start Autonomous Flight
- Flight Behavior Observed -

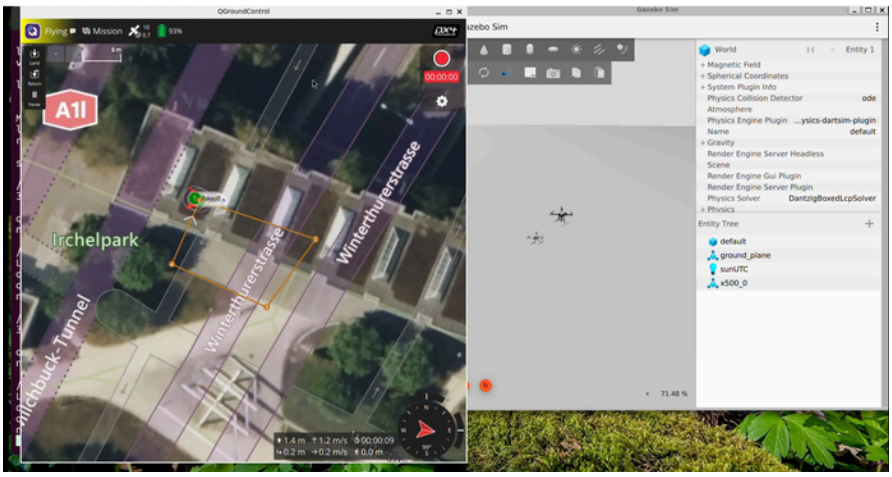
→During simulation: Drone takes off vertically, Reaches initial altitude, Moves towards first waypoint, Follows all waypoints sequentially, Adjusts altitude at each waypoint, Maintains stable flight path

- Return-to-Launch (RTL)-

→After completing mission: Drone automatically returns to home position, Lands safely

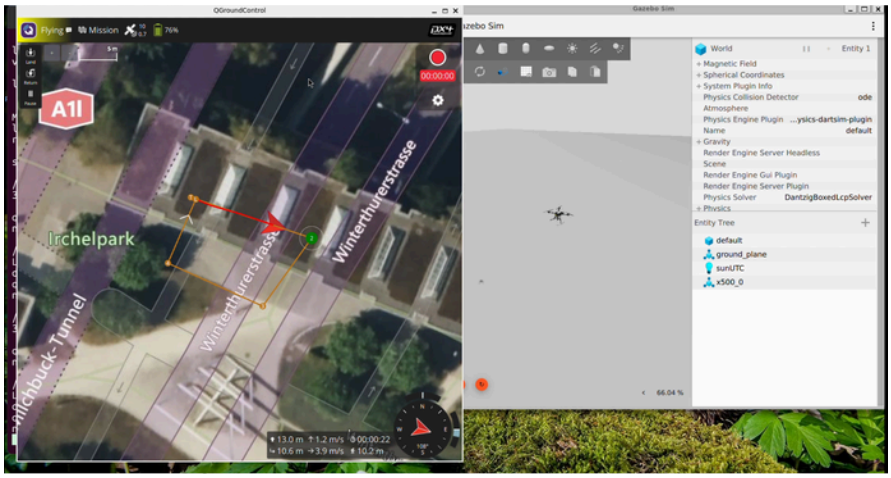
- Telemetry Observations-

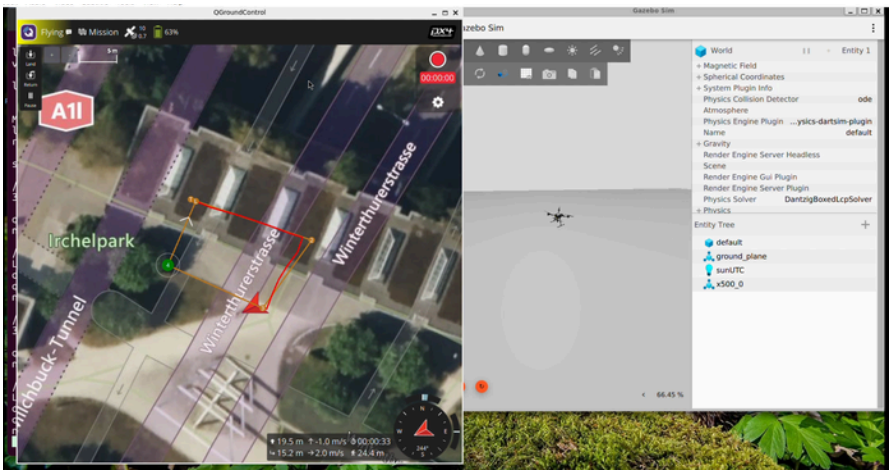
Parameter	**Observation**
Altitude	Changes as per waypoints
speed	stable during flight
path	slightly rough transitions
GPS	accurate within small error



TAKEOFF

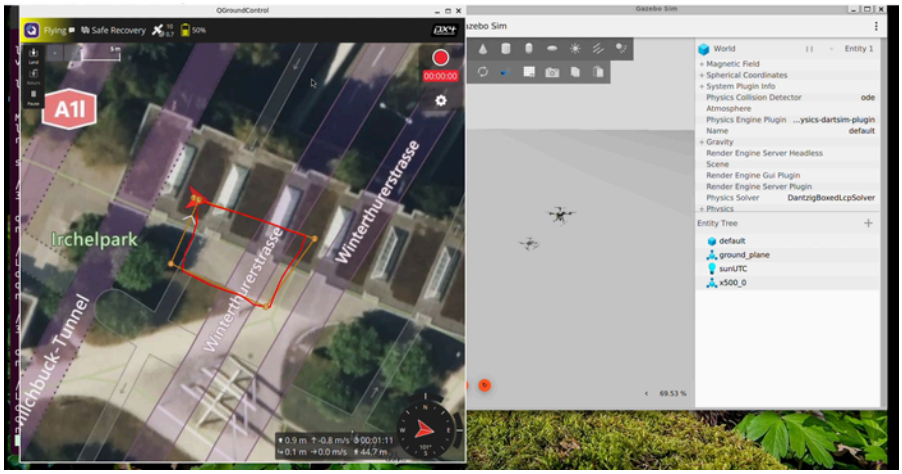
MID FLIGHT





TURNING
AT
WAYPOINT

LANDING



- Challenges Faced -
  → Terminal freezing during simulation
  → High system load (8GB RAM limitation)
  → Initial difficulty in switching to Auto Mode
- Solutions Applied -
  → Used gz_x500 instead of heavy Gazebo
  → Closed background applications
  → Restarted terminal when stuck
- Result - The autonomous mission was successfully executed. The UAV followed all waypoints accurately and completed the mission with return-to-launch functionality.

Part 4 : Understanding Navigation Logic

- Introduction- Navigation logic in UAVs defines how a drone decides its path, direction, and movement to reach a target location. In autonomous systems, this logic is handled by the flight controller such as PX4, which processes GPS data and executes waypoint missions through QGroundControl.
- Core Concept of Navigation Logic- Navigation logic works on three main inputs:
  → Position (GPS coordinates)
  → Target waypoint
  → Flight parameters (speed, altitude)
- Waypoint Navigation Logic -

1. Drone receives waypoint list
2. Takes off to defined altitude
3. Moves toward first waypoint
4. Checks position error
5. Corrects direction and speed
6. Repeats until all waypoints are completed

- Position Error Calculation - The drone calculates the difference between: Current position & Target waypoint . This difference is called error.
- Control System (PID Controller) -
  To reduce error, UAV uses: PID controller
  Components: P (Proportional): Corrects present error
  I (Integral): Fixes accumulated error
  D (Derivative): Predicts future error
  This ensures: Smooth flight, Stable motion & No oscillations
- Path Planning Logic - Path planning decides how the drone moves between waypoints.
  Types: Straight-line navigation
  Grid-based path
  Optimized paths (advanced)
- Navigation Modes in UAV - Manual Mode → Controlled by pilot
  Position Mode → Holds position
  Auto Mode → Executes mission

- Role of Sensors in Navigation -
  → Navigation logic uses multiple sensors: GPS → Position
  IMU → Orientation
  Compass → Direction
  Barometer → Altitude
- Communication System -
  → Drone and ground station communicate using: MAVLink
  → It sends: Commands, Telemetry data, Mission updates
- Navigation Flow (Simplified) - Receive mission → Takeoff → Navigate to waypoint → Adjust using PID → Reach target → Move to next waypoint → Return to launch
- Challenges in Navigation - GPS errors
  - Wind disturbances
  - Obstacle avoidance
  - Signal delay
- Conclusion - Navigation logic is the brain of autonomous UAV systems. By combining GPS data, control algorithms, and communication protocols, drones can perform precise and reliable missions without human intervention.

Part 5 : Industry level scenario - <u>Agricultural Field Survey using UAV</u>

- In real-world applications, autonomous drones are widely used in agriculture for monitoring large crop fields. A waypoint-based mission is created in a square or grid pattern covering the entire area. Similar to the simulation performed, the drone follows predefined GPS coordinates and maintains different altitudes at each waypoint. During the flight, it captures images and data related to crop health, irrigation, and soil conditions.
- The autonomous navigation system allows the drone to cover the field efficiently without manual control. It ensures accuracy, repeatability, and reduces human effort. After completing the mission, the drone returns to the launch point automatically using Return-to-Launch (RTL).This technique is part of precision agriculture and helps farmers improve productivity, reduce resource wastage, and monitor crops effectively.
- Benefits: → Reduces manual effort
  → Covers large areas quickly
  → Improves accuracy and efficiency
  → Safe operation in difficult environments
- The square waypoint path executed in the simulation is similar to real-world survey patterns used in agricultural drone missions.

Part6 : Industry Connection

- Delivery drones (Amazon-style logistics)
- Agricultural spraying and mapping
- Defense border surveillance
- Disaster management drones
- Infrastructure inspection

- Why Autonomous Navigation is Important

→ Autonomous navigation allows a UAV to perform missions without continuous human control by following pre-defined instructions and GPS-based waypoints. Modern drone systems using PX4 and ground stations like QGroundControl rely heavily on this capability.

→ Key Importance: -Precision & Accuracy
-Reduced Human Effort
-Scalability of Operations
-Consistency in Missions
-Access to Difficult Areas

- Benefits Over Manual Control: - Higher Efficiency
  - Improved Safety
  - Better Accuracy
  - Continuous Operation
  - Complex Mission Capability
  - Data Collection & Automation
- Conclusion - Autonomous navigation is a fundamental technology in modern UAV systems. Compared to manual control, it offers higher efficiency, accuracy, safety, and scalability, making it essential for real-world applications such as delivery, surveillance, and mapping.